5.3.16 Extended interrupt and event controller input (EXTI) characteristics

The pulse on the interrupt input must have a minimal length in order to guarantee that it is detected by the event controller.

Table 52. EXTI input characteristics

Specified by design and not tested in production.

Symbol	Parameter	Conditions	Min	Typ	Max	Unit
PLEC	Pulse length to event controller	-	20	-	-	ns

5.3.17 12-bit analog-to-digital converter ADC characteristics

Unless otherwise specified, the parameters given in Table 53 are values derived from tests performed under ambient temperature, f_{HCLK} frequency, and V_{DDA} supply voltage conditions summarized in Table 17.

Note: It is recommended to perform a calibration after each power-up.

Table 53. 12-bit ADC characteristics

Specified by design - Not tested in production.

Symbol	Parameter	Conditions	Min	Typ	Max	Unit
V_{DDA}	Analog power supply for ADC ON	-	2.70	-	3.6	V
V_{REF+}	Positive reference voltage	-	2.5	-	V_{DDA}	V
V_{REF-}	Negative reference voltage	-	V_{SSA}			V
f_{ADC}	ADC clock frequency	2.7V ≤ V_{DDA} ≤ 3.6 V	8	-	36	MHz
f_S with R_{AIN}=47 Ω and C_{PCB} = 22 pF	Sampling rate for slow channels	Resolution = 12 bits; All modes; 2.7 V ≤ V_{DDA} ≤ 3.6 V; -40°C ≤ T_J ≤ 140°C; f_{ADC}= 36 MHz; SMP = 2.5	-	2.25	-	MSPS
		Resolution = 10 bits	-	2.57	-	
		Resolution = 8 bits	-	3	-	
		Resolution = 6 bits	-	4.5	-	
t_{TRIG}	External trigger period	Resolution = 12 bits	16	-	-	$1/f_{ADC}$
V_{AIN}	Conversion voltage range	-	0	-	V_{REF+}	V
R_{AIN}(1)	External input impedance	Resolution = 12 bits, T_J = 140°C	-	-	110	Ω
		Resolution = 12 bits, T_J = 125°C	-	-	610	
		Resolution = 10 bits, T_J = 140°C	-	-	2305	
		Resolution = 10 bits, T_J = 125°C	-	-	4290	
		Resolution = 8 bits, T_J = 140°C	-	-	11110	
		Resolution = 8 bits, T_J = 125°C	-	-	15860	
		Resolution = 6 bits, T_J = 140°C	-	-	46890	
		Resolution = 6 bits, T_J = 125°C	-	-	79000	